

FNVXR Offline 6DoF Coordinate Proof

CURRENT VERDICT: NOT READY FOR ANOTHER HEADSET RUN

The spatial coordinate theorem and deterministic numeric tests pass after concrete repairs. The complete runtime claim does not pass because exact shader ownership and full eye-target draw classification remain incomplete. This report deliberately separates proven math from missing evidence.

Item	Result
Offline numeric proof	PASS
Exact-source 32-bit/64-bit build, CTest, safety audit	PASS
Print mesh digital validation	PASS
External slicer validation	NOT_RUN
Runtime shader/draw proof	FAIL - hard blockers remain
Headset used by this proof run	NO

1. Coordinate definitions

OpenXR local coordinates are +X right, +Y up, +Z back, so forward is -Z. A NiCamera local rotation stores columns (right, up, back), so OpenXR translation is already in the correct camera-local basis. Actor coordinates are different: (right, forward, up). Those two mappings must never be substituted for each other. The configured 70 units/m is an approximate FNV geometry scale. not a measured runtime calibration. See GECK Units.

```
XR_to_NiCamera_local = I
XR_to_actor * (x,y,z)^T = (x,-z,y)^T
s = configured 70 Fallout units/m (approximate geometry scale)
```

2. Rigid 6DoF theorem

Let p_o be the recenter eye midpoint, R_o the gravity-aligned yaw-only origin, C the leveled NiCamera body rotation. b the engine camera anchor. p_i/R_i an OpenXR eye pose. and X a room point.

```
d_i = R_o^T (p_i - p_o)
R_i_local = R_o^T R_i
p_i_game = b + s C d_i
R_i_game = C R_i_local
X_game = b + s C R_o^T (X - p_o)
```

Substitution and cancellation give:

```
R_i_game^T (X_game - p_i_game)
= (C R_o^T R_i)^T [s C R_o^T (X - p_o)]
= s R_i^T R_o C^T C R_o^T (X - p_o)
= s R_i^T (X - p_o)
```

Therefore every eye-local direction and perspective ratio is preserved. This is an exact similarity transform when all inputs belong to the same source pose transaction.

3. Numeric falsifiers and regressions

Check	Trials	Max error	Tolerance	Result
Both declared basis maps are proper rotations	2	0	1e-12	PASS

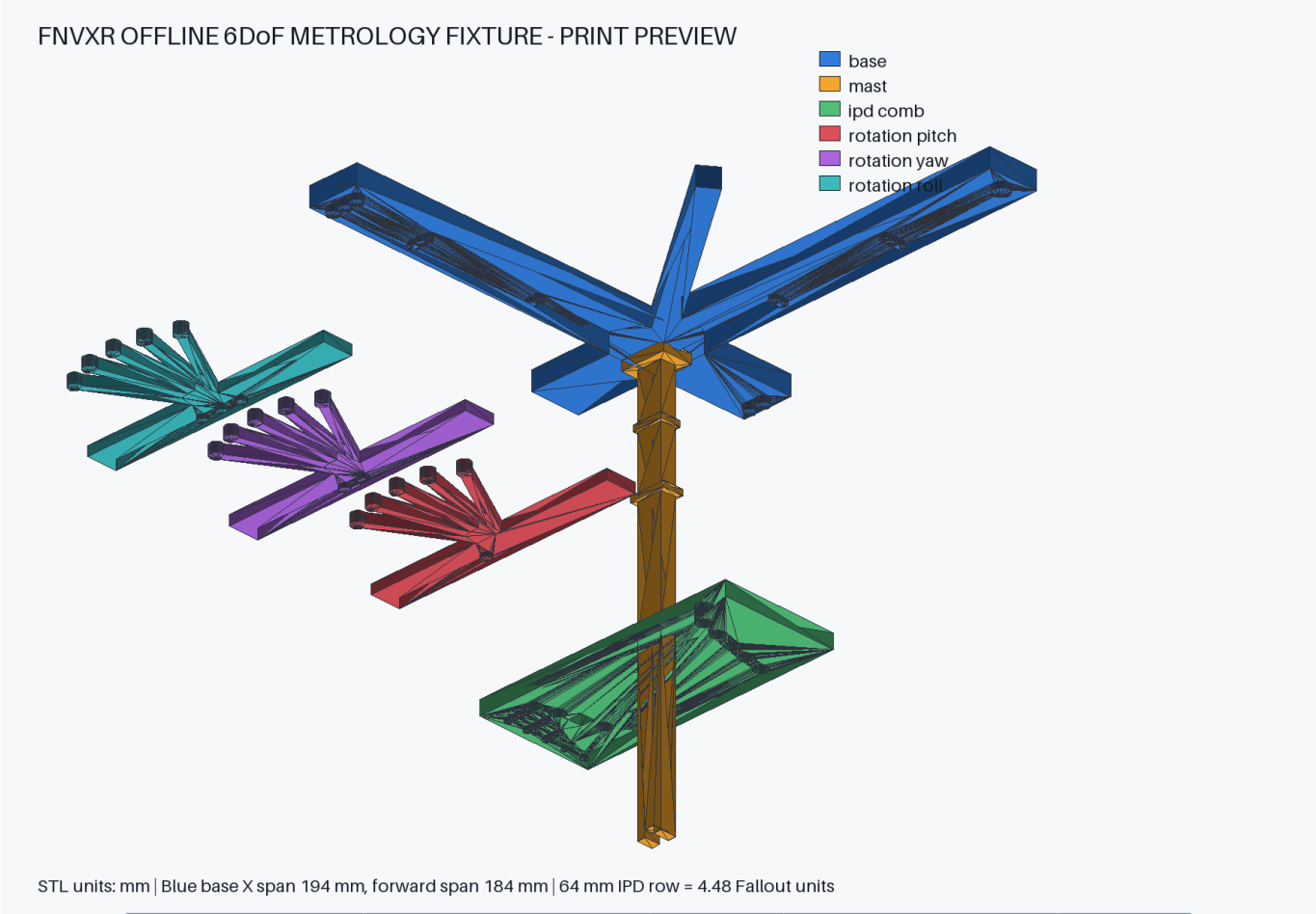
Check	Trials	Max error	Tolerance	Result
Rigid 6DoF eye-space equivalence	5000	1.71e-11	5e-09	PASS
NiCamera-local translation has no cross-axis leakage	3000	1.94e-16	1e-12	PASS
Recenter removes absolute height and preserves relative lift	2	6.22e-15	1e-12	PASS
64 mm eye baseline has correct signs and scale	1	0	1e-12	PASS
Yaw-only recenter removes gravity-axis mixing counterexample	1	7.44e-18	1e-12	PASS
Translated/canted eye frusta stay inside center traversal union	14400	0	1e-12	PASS
Double-precision VP delta reconstructs captured-scale WVP	501	4.99e-12	1e-07	PASS
Final patched-WVP gate detects draw-local float amplification	1	0	0	PASS
Source-pose age gate accepts 20 ms and rejects 100 ms/future 10 ms	3	0	0	PASS

Old axis counterexample: 100 mm physical forward is XR (0,0,-0.1). The removed actor permutation produced NiCamera-local [0.0, 0.1, 0.0], which is upward. A 100 mm physical lift became camera-back instead of up.

Old recenter counterexample: a 30 degree pitched origin conjugating a 90 degree world yaw gives forward [-0.866025, 0.433013, -0.25], including 0.433 vertical leakage.

4. Python-generated asymmetric print candidates

The fixture makes axis permutation, reflection, scale, IPD sign, relative height, and rotation-axis mistakes physically measurable. It cannot by itself prove what the runtime renders. STL is unitless; import every part as millimeters. These remain print candidates until an external slicer accepts them.



Part	mm size	Triangles	SHA-256 prefix	Checks
base	194.0 x 184.0 x 10.5	1644	845844d82e870e77	PASS
mast	18.0 x 12.0 x 181.0	156	3483573d2b069bc5	PASS
ipd comb	104.0 x 46.0 x 7.5	2308	350fa4177f52755c	PASS
rotation pitch	100.0 x 53.4 x 5.8	460	ec57198ec9d42e06	PASS
rotation yaw	100.0 x 53.4 x 5.8	538	4a736ad43a41351f	PASS
rotation roll	100.0 x 53.4 x 5.8	618	763a4ec4a52e6329	PASS

Digital gates: finite vertices, no degenerate/duplicate triangles, every edge incident to exactly two faces, one connected component, positive signed volume, trimesh watertight/winding checks, manifold3d NoError, no triangle edge below 0.05 mm, no triangle area below 0.001 mm², STL roundtrip bounds <= 0.02 mm, volume error <= 0.1%, and <= 210 mm XY envelope. Nominal design-feature sizes are declared, not measured wall-thickness results. External slicer validation is NOT RUN because no supported slicer is installed.

5. Five independent oracle reviews

Oracle	Verdict	Independent finding
1	CONDITIONAL PASS	Independently proved the rigid similarity transform and confirmed the native-axis, yaw-only recenter, 70-unit scale, displaced-eye cull, and exact source-pose repairs. Refused headset certification because no-depth translation needs a strict age bound and old logs cannot validate the new binary.
2	NOT READY	Independently reproduced the black screen from missing exact shader contracts: coverage fell to 391/392, the producer aged out, and disabled fallbacks left no visible gameplay layer. It also found incomplete oPos semantics, final-camera timing, stale fixed-state classification, and draw-ledger gaps. The first three implementation-level gaps were tightened afterward; missing bytecode and complete draw ownership remain red.
3	FIXTURE CONDITION	Specified an asymmetric/chiral physical metrology target, IPD comb, height stations, rotation witnesses, and independent mesh checks, while warning that a print cannot prove runtime pose or stereo. Its initial one-unit-per-inch scale assumption was rejected; the generated fixture uses the cited approximate 70 units/m scale.
4	FAIL END-TO-END	Confirmed the spatial theorem, eye/FOV gates, cull construction, source transaction, and post-traversal camera guard. It found final patched-WVP float amplification, configured-skip/fixed-function/UP draw denominator holes, weak skip identities, incomplete shader semantics, and recorded 89-311 ms source age. A double-reference final-WVP gate was added; draw ownership, shader corpus, and temporal feasibility remain blockers.
5	CORE PASS / PACKAGE FAIL	Independently repeated 20,000 rigid-transform cases, verified scale/IPD and every raw STL topology/hash, and judged the red PDF honest. It found float-intermediate frustum under-bounds, no mast hard stop/datum, Boolean slivers, asserted thickness, missing trace hashes, and pending review rows. The generator now uses double cull transforms, a compiled long-range test, a flange datum, simplified/sliver-gated meshes, nominal-feature wording, exact-source build/test hashes, and these five populated reviews. External slicing and runtime proof remain red.

6. Exact-source traceability

Gate	Exit	Output SHA-256 prefix	Result
build win32 release	0	af73849350fd0a83	PASS
ctest win32 release	0	a92e01abd1a28fd2	PASS
build x64 release	0	8a4d27eb01f4758d	PASS
ctest x64 release	0	7bf3875060d41137	PASS
launch safety audit	0	b6d3037f303907ba	PASS
shader contract dataflow selftest	0	c8ca19bda5653baf	PASS

File	Bytes	SHA-256 prefix
tools/generate_6dof_proof_artifacts.py	63867	18d480972def0676
docs/6dof-coordinate-proof.md	5160	8a89bc147b1285da
renderhook/fnvxr_stereo_math.h	32476	5c9cfdae6b9bef96
renderhook/fnvxr_d3d9_proxy.cpp	501667	49ac89822ac1ad30
host/fnvxr_openxr_pose_host.cpp	353242	baa0662b2af3d1ba
plugin/fnvxr_nvse_plugin.cpp	369036	69fe527d06d29e1f
tests/fnvxr_stereo_math_test.cpp	33667	767925416c51dc6a
scripts/get-verified-shader-wvp-contracts.ps1	13761	ae9b6765b68f6cfe
scripts/fnvxr-sidecar-common.ps1	59775	7492a7d92597156b

File	Bytes	SHA-256 prefix
scripts/audit-launch-safety.ps1	129109	387cf40a1db4c888
build-win32/Release/d3d9.dll	243200	439ceddc29d8420a
build-win32/Release/nvse_fnvvr.dll	147456	a88a3a68d3c0e62e
build-win32/Release/fnvvr_stereo_math_test.exe	42496	6af0d48be43c7d65
build/Release/fnvvr_openvr_pose_host.exe	246784	51e0a8b130636516
build/Release/fnvvr_stereo_math_test.exe	44544	139f74a1eb1ecdfe

7. Hard blockers - why the headset stays off

1. A historical run introduced vertex shaders 0x5999d5b0 and 0x83e14aae without verified WVP contracts. Coverage fell to 391/392, the producer retained then lost its last frame, and the host submitted no visible fallback layer. This deterministically explains the black screen.
2. The contract generator now requires a complete, unoverwritten oPos.xyzw provenance chain, and therefore rejects parts of the legacy 45-contract corpus. No replacement full contract set exists for all observed shader forms or the missing shaders.
3. The strict-eye-target draw ledger does not yet prove exactly one semantic classification for every contributing programmable, fixed-function, configured-skip, and composite draw.
4. Configured skip pairs use 32-bit FNV pairs rather than a strong SHA-256 plus byte-length semantic manifest.
5. A projection layer without depth cannot exactly reproject translated old pixels. The new 25 ms gate fails closed, but production performance inside that budget is not yet evidenced.
6. There is no new live binary evidence after these repairs. Existing logs are evidence of the old failures only.

8. Acceptance rule

Do not launch the headset from this proof package. The next eligible step is still offline: capture missing shader bytecode in producer-only mode with a visible monitor fallback, satisfy strict oPos semantics, produce a zero-unclassified-draw ledger, pass external slicing, and regenerate this report with all red gates green. Only then is one short instrumented headset validation justified.

Regeneration: use the bundled Python runtime to run tools/generate_6dof_proof_artifacts.py. Deterministic numeric seed: 1829707814. Numeric trials: 22911.